EE4047 – High Voltage Techniques I

Continuation to:
Generation of High Voltages

Week 5.1– Monday Lesson

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OUTLINE

3 Continuation to: Alternating Voltages

3.2 Series resonant circuits

2.2 Series resonant circuits

The tuned series resonant HV testing circuit is developed for overcoming the accidental and unwanted resonance to which more conventional test sets are more prone.

If we consider a conventional straight test set system consisting of a transformer, testing a capacitor C, then its equivalent circuit will be $(r_1 + j\omega L_1)$ representing the impedance of the voltage regulator and the transformer primary, ωL represents the transformer shunt impedance, which is usually large compared with L_1 and L_2 and can be normally neglected. $(r_2 + j\omega L_2)$ represents the impedance of the transformer secondary. $1/\omega C$ represents the impedance of the load.

If by chance $\omega(L_1 + L_2) = 1/\omega C$, accidental resonance occurs. At supply frequency, the effect can be extremely dangerous as the instantaneous voltage application can be of the order of 20 times the intended high voltage.

This is highly possible to occur when testing at the maximum limit of current and relatively low voltage, i.e. High capacitive load.

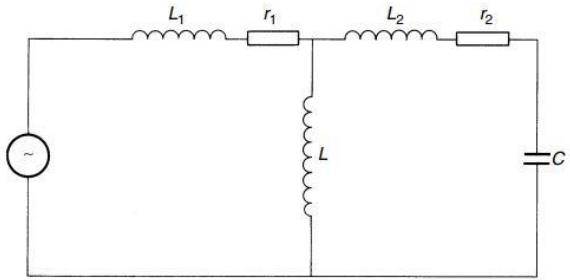


Figure 2.17 *Equivalent circuit of straight test set consisting of a transformer and test capacitance*

2.2 Series resonant circuits

With the series resonant set, however, the resonance is controlled at fundamental frequency and no unwanted resonance can therefore occur.

In each circuit shown in the figure, the capacitance C_t represents the almost pure capacitive load of the test objects and a constant power supply frequency is assumed.

In subfigure (a), a continuously variable inductance (reactor) is connected to the LV winding of a step-up trafo, whose secondary winding is rated for the full test voltage. In this way, the impedance of the reactor is converted to the HV side. If the inductance of the reactor is tuned to match the impedance of the capacitive load, the reactive power of the load is completely compensated.The step-up transformer, however, has to carry the full load current which is a disadvantage of this circuit.

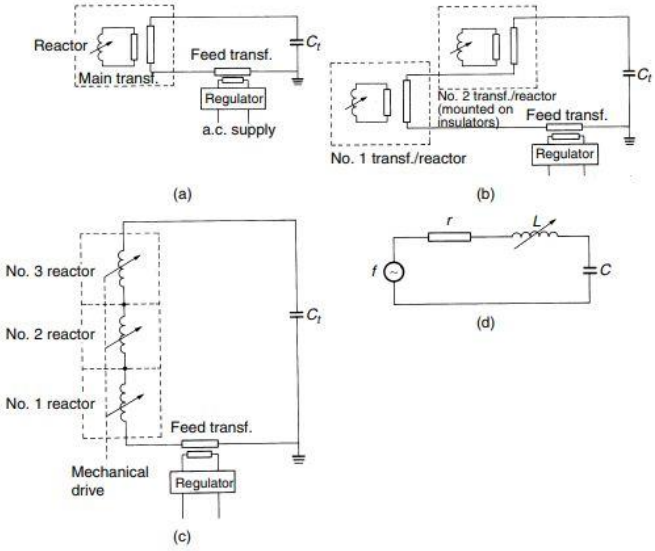


Figure 2.18 *Series resonant circuit for transformer/reactor. (a) Single transformer/reactor. (b) Two or more units in series. (c) Simplified diagram of s.r. circuit for h.t. reactor units in series. (d) Equivalent diagram of s.r. circuits*

The same disadvantage applies to circuit in subfigure (b), although no special means are necessary to cascade two or more units. The inductors are designed for high-quality factors $Q = \omega L/R$ within the limits of the inductance variation.

In subfigure (c), the inductance of the reactors may be changed by up to 10 to 20 times, offering the opportunity to tune the circuits with capacitances which vary within the same order.

The equivalent circuits for all these options is simply a low damped circuit shown in subfigure (d).

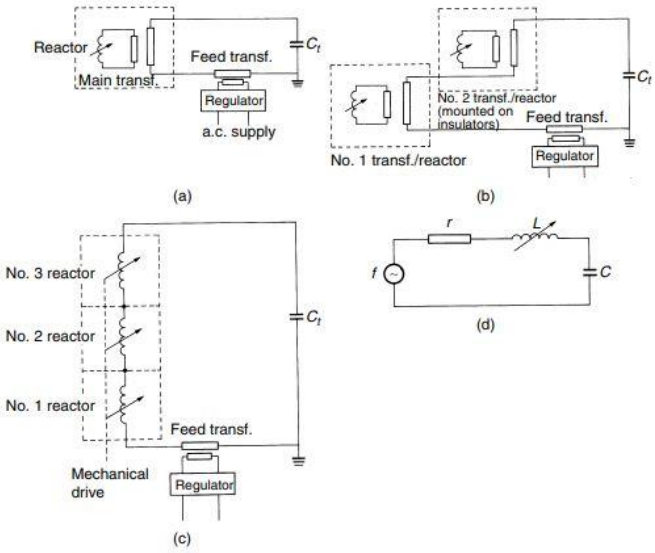


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3 ALTERNATING VOLTAGES

Additional advantages of the series resonant circuits:

- Voltage waveshape is improved not only by the elimination of unwanted resonances, but also by attenuation of harmonics already in the power supply.

- The power required from the supply is lower than the kVA in the main test circuit. (only 5% of the main kVA with a unity power factor).

- If a failure of the test specimen occurs, no heavy power arc will develop as only the load capacitance will be discharged. Additionally the arc will be self-extinguishing due to voltage collapse.
- The series or parallel operation of the HV reactor or LV reactor and HV transformer units is simple and very efficient.
- Auto-tuning devices can keep the set in tune, if supply frequency or load capacitance varies, or concerning auto-voltage control.

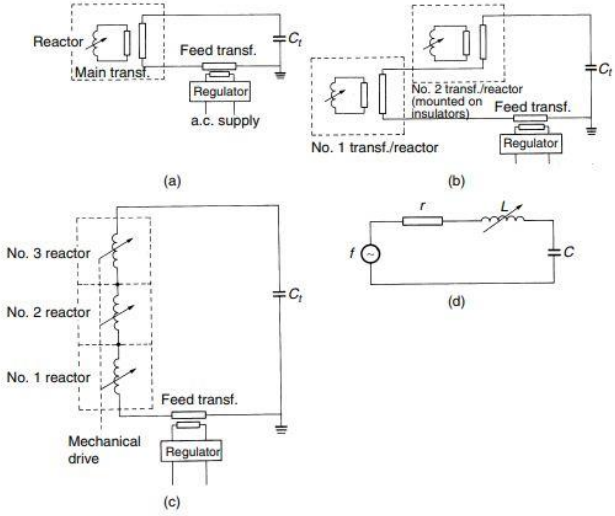


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As the load capacitance cannot be kept within narrow limits, the supply frequency can be continuously variable to achieve resonance. As shown in the figure, an exciter circuit connected to the LV mains,can be used to excite the circuit with a variable frequency, this supply is therefore designed as a controlled frequency converter, which are standard devices nowadays. This converter supplies again only the losses of the testing circuit, which are usually less than about 2% of the reactive power of C_t if frequencies equal to or higher than 50 Hz are used.

Due to resonance condition the frequency is always

$$f = \frac{1}{2\pi\sqrt{L_n C_t}}$$

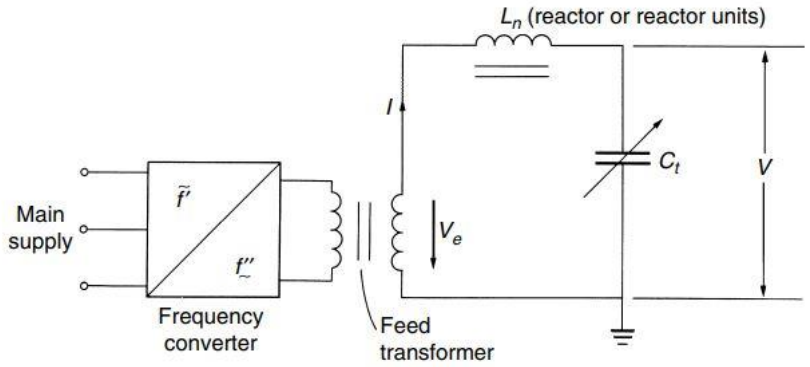


Figure 2.20 *Schematic diagram of s.r. test circuit with variable test frequency*

The nominal inductance L_n will be designed according to a nominal capacitance $C_n = C_t$ which is the highest capacitance that can be tested with the full rated voltage $V = V_n$ of the circuit and a nominal frequency f_n, which is the lowest frequency within this rated voltage.

$$L_n = \frac{1}{(2\pi)^2 f_n^2 C_n}$$

A further criterion for the design is the maximum or nominal current $I = I_n$, which either overheats the coil or saturates the iron core, if any. As the losses are very small, we may neglect $R \ll \omega L_n$.

I_n may directly be derived from the voltage drop across L_n

$$I_n = \frac{V_n}{2\pi f_n L_n} = V_n \sqrt{\frac{C_n}{L_n}}$$

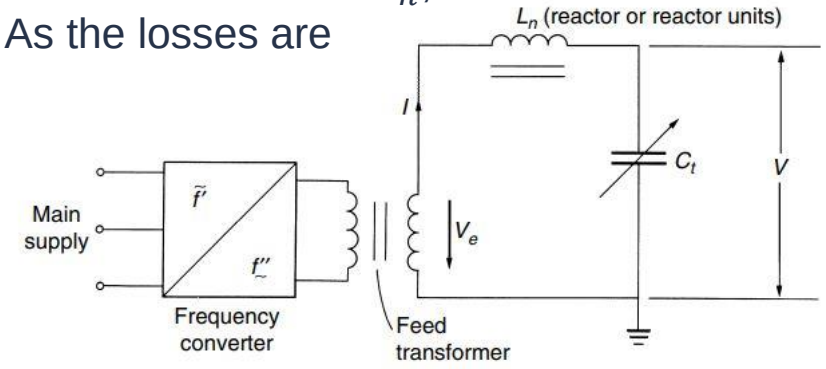


Figure 2.20 *Schematic diagram of s.r. test circuit with variable test frequency*

For test objects with capacitance values C_t different from C_n, the resulting testing frequency f will also be different from f_n. It can be calculated as

$$\frac{f}{f_n} = \sqrt{\frac{C_n}{C_t}} = \frac{1}{\sqrt{C_t/C_n}}$$

The below relation is present

$$\frac{I}{I_n} = \frac{f_n}{f} = \sqrt{\frac{C_t}{C_n}}$$

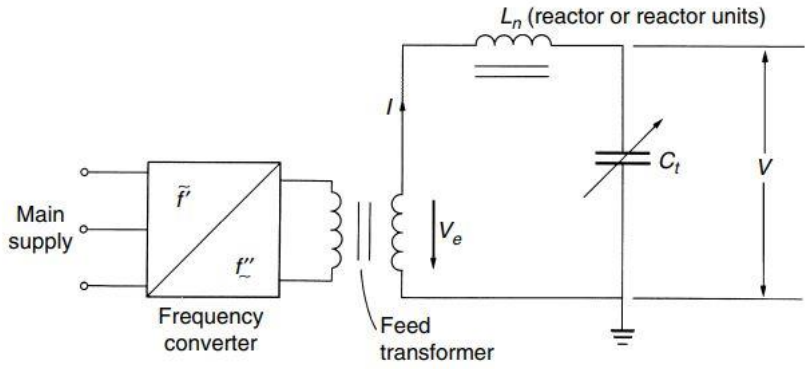


Figure 2.20 *Schematic diagram of s.r. test circuit with variable test frequency*

The normalized operating conditions are illustrated in the figure. For quite small test specimens, the test frequency f may conveniently be limited by the addition of a permanent capacitor as e.g. a capactive voltage divider; the relatively modest variation of this frequency for large capacitors under test will improve the flexibility of applications, i.e. for the testing of power cables with AC voltages.

The actual limitations in testing of very large test specimens with lower voltages than V_n are given by the reduction of Q for too low frequencies, and the frequency for which the exciter transformer saturates.

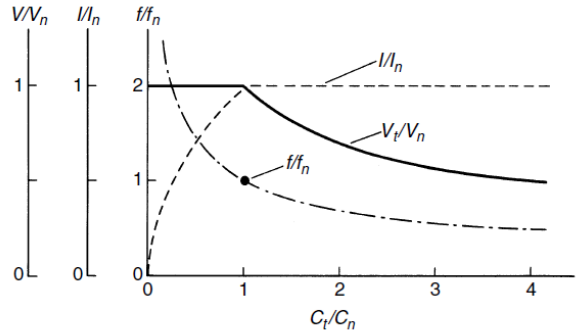


Figure 2.21 *Operating characteristics of circuit according to Fig. 2.20*

END OF THE LESSON

Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm